

Naive Bayes Foundations: A Bayes' Theorem Problem

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Where this fits: Pipeline map step 0 (Foundations). See the [Course map](#).



- **Builds on:** Conditional probability ([Note MA-015](#)); Joint and marginal probability ([Note MA-014](#)).
- **Leads to:** Naive Bayes ([Note ML-081](#)); MAP estimation ([Note MA-072](#)); Gaussian mixture model (GMM) ([Note MA-073](#)).

1. Overview

Key point: Three machines make markers; a random marker turns out defective. Bayes' theorem, with the total probability rule for the evidence, gives the chance it came from each machine.

This Note works through a classic problem with Bayes' theorem. The Note also introduces the **law of total probability** (G-1053), the usual way to compute the evidence $P(B)$ in the denominator. Naive Bayes computes its evidence the same way.

2. The problem

Key point: Production shares 20%, 30%, 50%; defect rates 5%, 3%, 1%.

A factory makes markers on three machines:

Machine	Share of all markers	Defect rate of that machine
M1	20%	5%
M2	30%	3%
M3	50%	1%

A marker is picked at random from a batch and turns out to be **defective**. What is the probability that it was made by M3?

Figure 1 shows the idea before any formula. In a typical batch of 1000 markers, M1, M2 and M3 make 200, 300 and 500. Their defect rates leave 10, 9 and 5 defective markers. Knowing the picked marker is defective means it is one of these 24, and only 5 of the 24 came from M3.

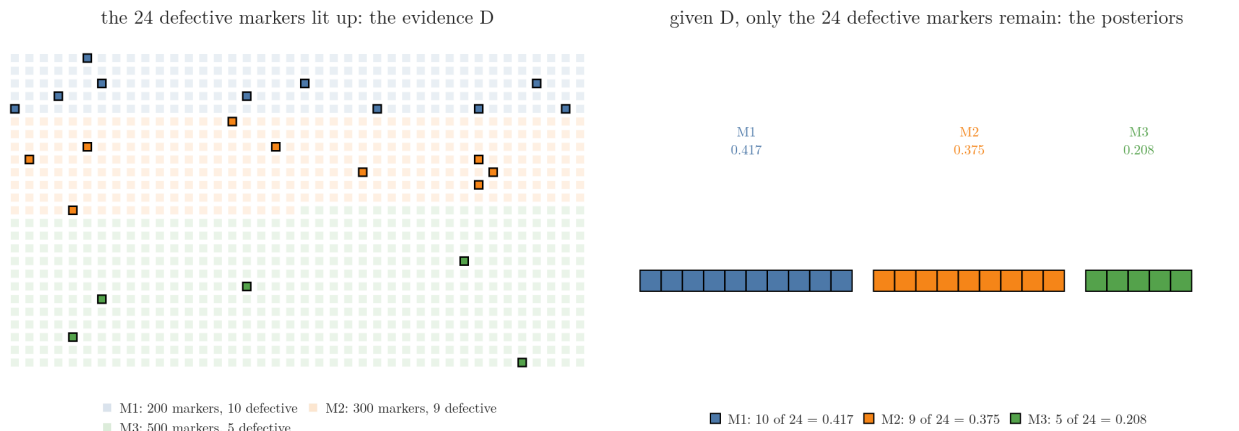


Figure 1: A batch of 1000 markers, one square each, coloured by machine. The 24 defective ones light up (10 from M1, 9 from M2, 5 from M3); given that the marker is defective, only those 24 remain, and M3 supplies 5 of 24 = 0.208.

3. Writing down what we know

Key point: The production shares are priors $P(M)$; the defect rates are likelihoods $P(D | M)$.

Let D be “the marker is defective”. In probability language:

- **Priors** (G-1565): $P(M1) = 0.2$, $P(M2) = 0.3$, $P(M3) = 0.5$.
- **Likelihoods** (G-1086): $P(D | M1) = 0.05$, $P(D | M2) = 0.03$, $P(D | M3) = 0.01$.

The question asks for $P(M3 | D)$. By **Bayes’ theorem** (G-269),

$$P(M3 | D) = \frac{P(D | M3) \times P(M3)}{P(D)}$$

Everything on the right is known except $P(D)$, the overall probability that a marker is defective.

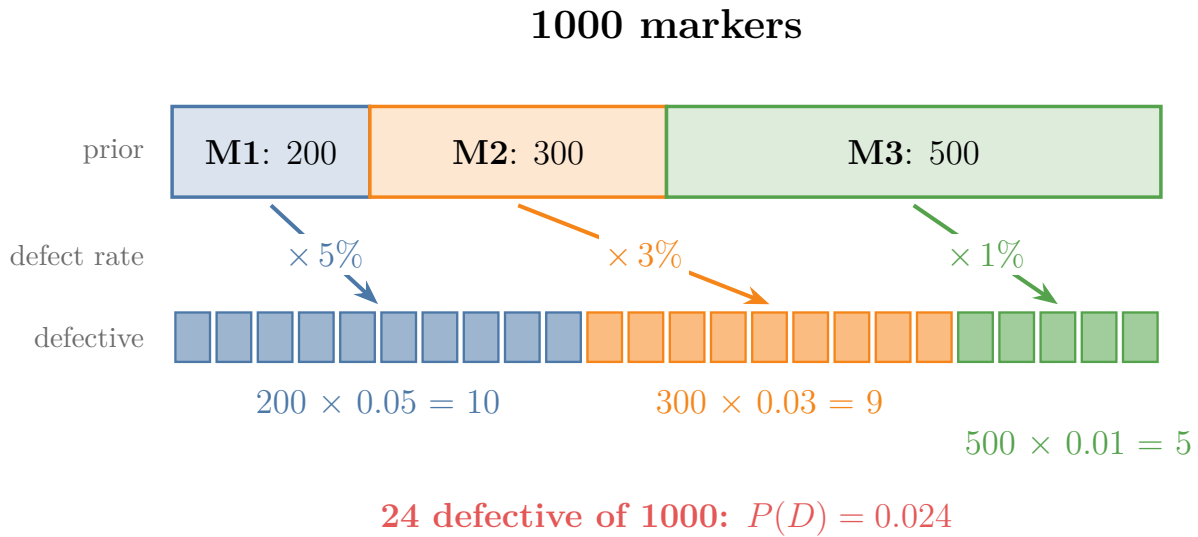


Figure 2: The problem as counts in a batch of 1000 markers. Top: the priors, with widths to scale. Bottom: one square per defective marker, from each machine's count times its defect rate

Figure 2 turns the priors and likelihoods into expected counts for 1000 markers. Watch the bottom row: M3 makes the most markers but supplies the fewest defective squares.

4. The evidence: total probability

Key point: A defective marker came from exactly one machine, so $P(D)$ is the sum of the three “defective and from machine i ” probabilities.

A defective marker must have come from M1, M2 or M3, and from only one of them. These three cases are mutually exclusive (the mutually exclusive events Note), so their probabilities add:

$$P(D) = P(D \cap M1) + P(D \cap M2) + P(D \cap M3)$$

Each term is a likelihood times a prior, by $P(A \cap B) = P(B | A)P(A)$ from the Bayes' theorem proof:

$$P(D) = P(D | M1)P(M1) + P(D | M2)P(M2) + P(D | M3)P(M3)$$

This sum is the law of total probability. With numbers:

$$P(D) = 0.05 \times 0.2 + 0.03 \times 0.3 + 0.01 \times 0.5$$

$$P(D) = 0.010 + 0.009 + 0.005$$

$$P(D) = 0.024$$

So 2.4% of all markers are defective. Figure 3 shows the same calculation as a **probability tree** (G-1573).

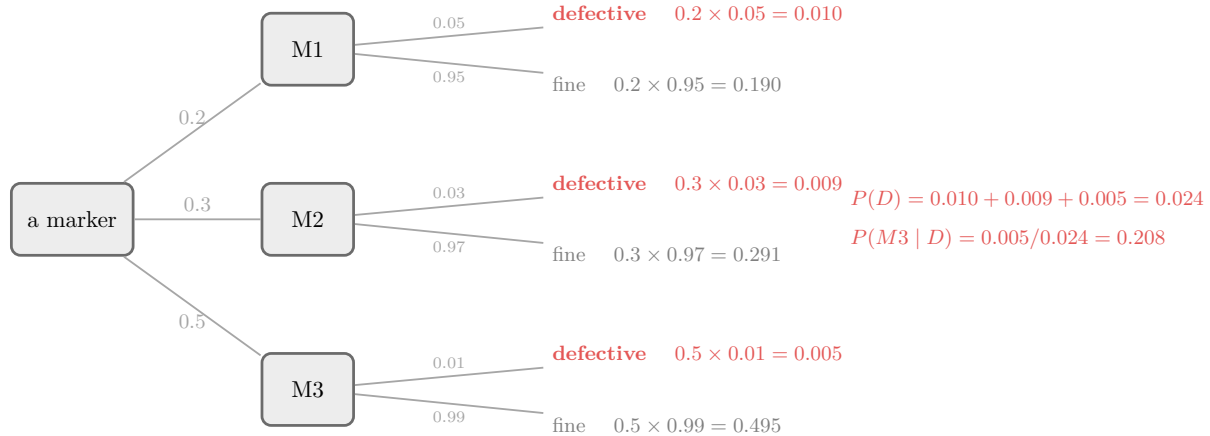


Figure 3: Probability tree: each path multiplies its branches

5. The answer

Key point: $P(M3 | D) = 0.005 / 0.024 = 0.208$. M3 makes half the markers but only about a fifth of the defective ones.

$$P(M3 | D) = \frac{0.01 \times 0.5}{0.024} = \frac{0.005}{0.024} = 0.208$$

The same calculation for every machine:

Machine	Prior $P(M)$	Joint probability (G-986) $P(D \cap M)$	Posterior (G-1536) $P(M D)$
M1	0.20	0.010	0.417
M2	0.30	0.009	0.375
M3	0.50	0.005	0.208

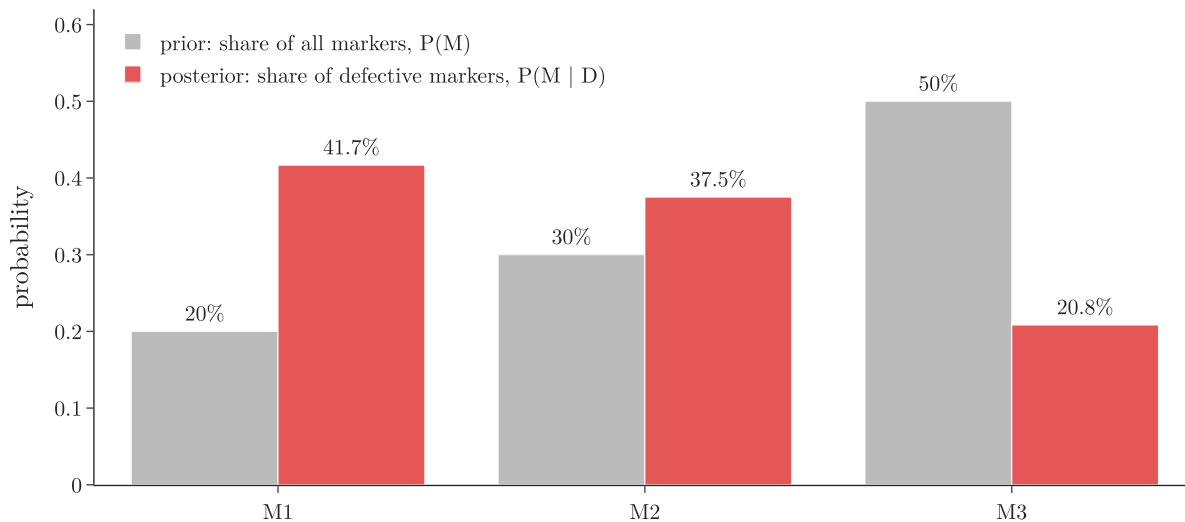


Figure 4: Share of all markers against share of defective markers

Seeing that the marker is defective changes the picture completely (Figure 4). Before, M3 was the most likely source (50%). After, M3 is the least likely (21%), because M3 rarely makes defects.

An everyday picture: three cooks share a kitchen, and one dish comes out burnt. The cook who makes the most dishes is not the likely culprit if that cook almost never burns anything. M1 makes only 20% of markers but 42% of the defective ones.

The three posteriors add up to 1, as they must: the marker came from some machine. A simulation of a million markers gives $P(M3 | D) = 0.2095$, matching 0.208.

Python: The whole calculation.

```
prior = {"M1": 0.2, "M2": 0.3, "M3": 0.5}
p_defect = {"M1": 0.05, "M2": 0.03, "M3": 0.01}

joint = {m: prior[m] * p_defect[m] for m in prior}
p_D = sum(joint.values()) # 0.024
{m: joint[m] / p_D for m in prior} # 0.417, 0.375, 0.208
```

6. Another way to see it: a medical test

Key point: A test does not give the final answer; it updates the prior. Posterior odds = prior odds $\times$ Bayes factor.

The marker problem has the same shape as a famous question about medical tests. The machines become “has cancer” and “does not have cancer”; “defective” becomes “the test is positive”.

A screening test is given to 1000 women. What we know:

- **Prior:** 1 percent have breast cancer, so 10 women are sick and 990 are healthy.
- **Sensitivity** (G-1641; recall, [Note ML-076](#)): the test is positive for 90 percent of the sick women: $P(+ \mid \text{cancer}) = 0.9$.
- **False positive rate** (G-749): the test is also positive for 9 percent of the healthy women: $P(+ \mid \text{healthy}) = 0.09$.

A woman tests positive. What is the chance that she has cancer? The test sounds “90 percent accurate”, so many people answer 9 in 10. Count instead, exactly as with the markers:

1. **Sick and positive** (true positives):

$$10 \times 0.9 = 9 \text{ women}$$

2. **Healthy and positive:** $990 \times 0.09 \approx 89$ women (false positives).

3. **All positives** (the evidence, by total probability):

$$9 + 89 = 98$$

4. **Posterior:** the share of positives who are sick.

$$P(\text{cancer} \mid +) = 9/98 \approx 0.092$$

That is about 1 in 11.

The healthy group is so large that its few false positives (89) swamp the true positives (9). This posterior is the **precision** (G-1547) of the test; in medicine it is called the positive predictive value.

The lesson: the test did not decide anything. The test moved the chance from 1 in 100 (the prior) to about 1 in 11 (the posterior). A test **updates** the prior.

6.1 The size of the update: the Bayes factor

Key point: Bayes factor = $P(\text{evidence} \mid \text{yes}) / P(\text{evidence} \mid \text{no})$. Multiply the prior odds by it to get the posterior odds.

How strong is the update? A positive result is 10 times more likely for a sick woman than for a healthy one:

$$0.9/0.09 = 10$$

This ratio of the two likelihoods is the **Bayes factor** (G-2215).

The Bayes factor works on **odds** (G-1376): the number of “yes” cases to the number of “no” cases.

1. **Prior odds:** 10 sick to 990 healthy, that is 1 : 99.
2. **Multiply by the Bayes factor.** These are the posterior odds.

$$1 \times 10 : 99 = 10 : 99$$

3. **Back to a probability:** 10 sick for every 99 healthy.

$$10/(10 + 99) = 0.092$$

The same answer as the count.

The rule is exact, and it follows from the count above: among the positives, the sick are scaled by 0.9 and the healthy by 0.09, so their ratio is multiplied by 0.9/0.09.

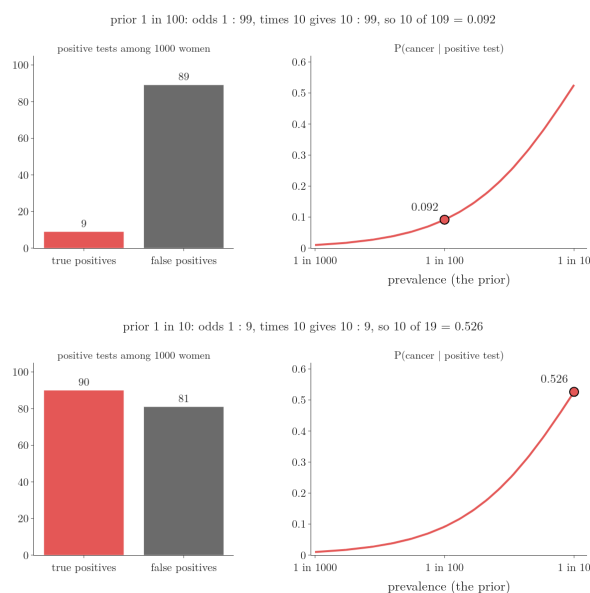


Figure 5: A test with sensitivity 0.9 and false positive rate 0.09 (Bayes factor 10), for priors from 1 in 1000 to 1 in 10. Left: expected true and false positives among 1000 women. Right: the posterior $P(\text{cancer} \mid \text{positive})$. The title gives the update in odds.

In Figure 5, watch the same test give very different answers as the prior changes. At a prior of 1 in 1000 a positive result means about 1 percent; at 1 in 100, 9 percent; at 1 in 10, the odds 1 : 9 become 10 : 9, which is 53 percent. The test is the same each time; only the prior differs.

The Bayes factor also previews Naive Bayes: each feature multiplies the score of a class by its own likelihood, one factor per feature (Note ML-081).

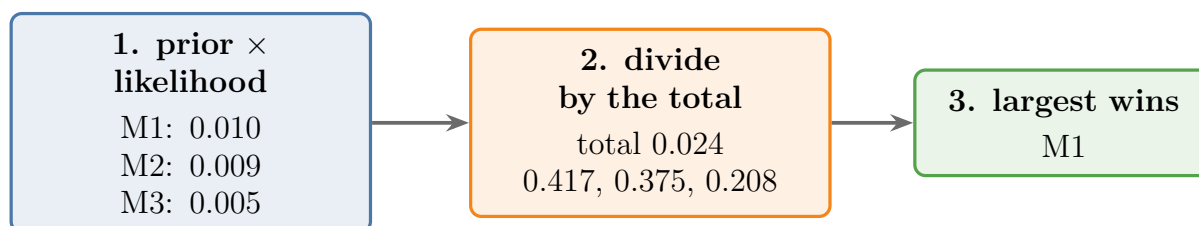
7. The pattern for Naive Bayes

Key point: For each class: $\text{prior} \times \text{likelihood}$. Divide by their total. The largest posterior wins.

A Naive Bayes classifier follows exactly this procedure. The machines become the classes, the values of the **target** (G-1949; the output we predict). “Defective” becomes the observed **features** (G-772; the input variables, one column each of the data table):

1. for every class, multiply its prior by the likelihood of the observed features;
2. divide each product by their sum (the evidence) to get posteriors that add to 1;
3. predict the class with the largest posterior.

Since the evidence is the same for every class, step 2 does not change which class wins; it only turns the scores into probabilities.



step 2 rescales every score by the same number: the winner does not change

Figure 6: The three Naive Bayes steps on the marker problem

Figure 6 runs the three steps on the marker numbers: M1 has the largest score both before and after dividing by 0.024.

8. Summary

- Priors: production shares; likelihoods: defect rates; posterior: which machine, given a defect.
- Law of total probability: $P(D) = \sum_i P(D | M_i)P(M_i)$. The sign $\sum_i$ means “add the terms for $i = 1, 2, 3$ ”, one per machine (0.010, 0.009 and 0.005, which add to 0.024).
- Answer: $P(M3 | D) = 0.208$; M1 is the most likely source (0.417).
- Medical test: prior 1 in 100, sensitivity 0.9, false positive rate 0.09 give a posterior of only $9/98 = 0.092$. Posterior odds = prior odds $\times$ Bayes factor (1 : 99 times 10 is 10 : 99).

9. Sources

Built from

- CampusX, “Naive Bayes Classifier | Part 5 | Problem based upon Bayes Theorem”, YouTube, <https://www.youtube.com/watch?v=aAEHjXDHtbE>
- Sanderson, G. (3Blue1Brown), “The medical test paradox, and redesigning Bayes’ rule”, YouTube, <https://www.youtube.com/watch?v=lG4VkPoG3ko> (the 1000-women example, the test as an update, the Bayes factor and the odds form; Figure 5 is our own chart of those numbers)

10. Key terms

Term	Meaning
Law of total probability	$P(B) = \sum_i P(B A_i)P(A_i)$, when the A_i are mutually exclusive and cover every case
Probability tree	A diagram in which each path multiplies the probabilities along its branches
Bayes factor (G-2215)	The ratio of the two likelihoods, $P(\text{evidence} \text{yes})/P(\text{evidence} \text{no})$; it multiplies the prior odds to give the posterior odds
Odds	The number of “yes” cases to the number of “no” cases, such as 1 : 99
False positive rate (G-749)	The share of real negatives that test positive
Joint probability	The probability that two events happen together, $P(A \cap B)$